Stage 1 Hard Drill — With Solutions

Chapter 3.2 — Market Microstructure & Order Book Fair Value

10 new competition-grade questions, full-chapter coverage, full worked solutions. Target: under 90 seconds each once the shortcut is spotted.

Q1 — Independence Shortcut on a Biased Die

Two independent biased dice each show $1, 2, 3, 4$ with $\mathbb{P}(k) = k/10$. A contract pays their *product* in dollars. Find the fair value.

Solution. First, $\mathbb{E}[k] = \sum_{k=1}^{4} k \cdot \frac{k}{10} = \frac{1+4+9+16}{10} = \frac{30}{10} = 3$. By independence, $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$:

$$\mathbb{E}[\text{payoff}] = 3 \times 3 = \boxed{\$9.00}$$

⋆ *Intuition:* never expand a 2-die joint table when the dice are independent — compute each marginal mean once, multiply. This is Example 1/2's linearity trick extended to products under independence.

Q2 — A Deliberate Trap: Does a Fair Value Even Exist?

A fair coin is flipped until the first heads, on flip k ($k = 1, 2, 3, \ldots$, so $\mathbb{P}(k) = 0.5^k$). A contract pays $\$2^k$. Find the fair value.

Solution.

$$\mathbb{E}[\text{payoff}] = \sum_{k=1}^{\infty} 2^k \cdot 0.5^k = \sum_{k=1}^{\infty} 1 = \boxed{\infty}$$

The sum diverges — **there is no finite fair value.** This is the classical *St. Petersburg paradox*.

⋆ *Intuition:* not every contract has a well-defined fair value just because you can write down $\sum$(payoff × probability) — a market maker facing this contract must cap exposure (a max payout, a position limit) rather than naively quote "half of infinity." Recognizing divergence *is* the correct Stage 1 answer, not a numerical approximation.

Q3 — Symmetry Beats Summation

A fair spinner shows integers 1 through 5 with equal probability. A contract pays $\$(6-k)$ if the spinner shows k. Find the fair value with no summation.

Solution. As k ranges uniformly over $\{1, \ldots, 5\}$, so does $6-k$ (it's just the same set reversed: $5, 4, 3, 2, 1$). So $6-k$ has the *same* mean as k itself:

$$\mathbb{E}[6-k] = 6 - \mathbb{E}[k] = 6 - 3 = \boxed{\$3.00}$$

▶ *Speed trick:* whenever a payoff is an affine transform of a uniform (or any symmetric) random variable, apply linearity to the transform directly — $\mathbb{E}[6-k] = 6 - \mathbb{E}[k]$ — instead of re-deriving the distribution of $6-k$ from scratch.

Q4 — Reverse-Engineering σ from a Quote

You observe a quoted market of bid \$97.80 / ask \$102.20. House convention: half-spread $= 1.1\sigma$. Recover the market maker's fair-value estimate and their assumed σ.

Solution. Fair value = mid $= \dfrac{97.80 + 102.20}{2} = \boxed{\$100.00}$. Half-spread $= \dfrac{102.20 - 97.80}{2} = 2.20$. Solve $1.1\sigma = 2.20$:

$$\sigma = \boxed{\$2.00}$$

⋆ *Intuition:* a quote is two numbers but encodes three things (fair value, spread convention, σ) — given any two, the house convention formula recovers the third. Reading a competitor's quote this way ("how uncertain do they think they are?") is a real trading skill, not just algebra practice.

Q5 — A Crossed Market

MM1 quotes bid \$50.30 / ask \$50.60. MM2 quotes bid \$50.00 / ask \$50.10. Explain why this market is "crossed," and find the risk-free arbitrage profit per unit.

Solution. Best bid across makers = \$50.30 (MM1); best ask across makers = \$50.10 (MM2). Since best bid > best ask, the market is **crossed**. Buy from MM2 at \$50.10, immediately sell to MM1 at \$50.30:

$$\text{profit} = 50.30 - 50.10 = \boxed{\$0.20 \text{ per unit, risk-free}}$$

▶ *Intuition:* in a liquid market this is instantly arbitraged away (both quotes get hit immediately) — a crossed market you can actually observe and act on is a brief, valuable anomaly, and spotting "best bid > best ask" at a glance is exactly the Stage 1 skill being tested.

Q6 — Fill Probability, Different Slope

A steeper fill-probability model: $\mathbb{P}(\text{fill}) = 0.6 - 0.3d$ (floored at 0), where d is distance from fair value in dollars. You place your ask at $d = 1.5$. Find $\mathbb{P}(\text{fill})$.

Solution.

$$\mathbb{P}(\text{fill}) = 0.6 - 0.3(1.5) = 0.6 - 0.45 = \boxed{0.15}$$

⋆ *Intuition:* a steeper slope (0.3 here vs. 0.25 in the chapter's base model) means fill probability decays *faster* with distance — this maker needs to quote closer to fair value to get filled at all, a sign of a more competitive or more liquid market.

Q7 — Asymmetric Fills After a Skew, Netted

After skewing for a long position: $\mathbb{P}(\text{fill on bid}) = 0.1$ (edge \$1.20 if filled), $\mathbb{P}(\text{fill on ask}) = 0.4$ (edge \$0.80 if filled). You hold 6 units, costing \$0.03/unit/round. Find the net expected P&L.

Solution. Gross trading P&L:

$$\mathbb{E}[\text{P\&L}] = 0.1(1.20) + 0.4(0.80) = 0.12 + 0.32 = 0.44$$

Inventory cost: $0.03 \times 6 = 0.18$. Net:

$$0.44 - 0.18 = \boxed{\$0.26}$$

▶ *Speed trick:* compute gross P&L as $\sum(\text{edge}_i \times \mathbb{P}(\text{fill}_i))$ exactly as in the chapter's base example, then subtract holding cost *last*, as its own separate term — never blend it into either side's edge.

Q8 — Reverse-Engineering Inventory from the Reservation Price

Fair value \$200. A market maker's reservation price is observed at \$194. Known: $\gamma = 0.02$, $\sigma^2 = 50$. Recover their inventory position q.

Solution. $r = \text{FV} - q\gamma\sigma^2 \Rightarrow 194 = 200 - q(0.02)(50) = 200 - q$

$$q = 200 - 194 = \boxed{6 \text{ units long}}$$

⋆ *Intuition:* just like Q4 for quotes, the reservation-price formula can be read backwards — watching *how far* a competitor's effective center has drifted from the market's fair value tells you their inventory sign and rough size, without ever seeing their book.

Q9 — Capstone: Full Pipeline from Scratch

A contract pays \$20/\$80/\$140 w.p. 0.4/0.4/0.2. House convention: half-spread $= 0.015\sigma$. Find the fair value, σ, and the quoted market.

Solution. $\mathbb{E}[X] = 20(0.4) + 80(0.4) + 140(0.2) = 8 + 32 + 28 = \boxed{\$68}$. $\mathbb{E}[X^2] = 400(0.4) + 6400(0.4) + 19600(0.2) = 160 + 2560 + 3920 = 6640$. Var $= 6640 - 68^2 = 6640 - 4624 = 2016 \Rightarrow \sigma = \sqrt{2016} \approx 44.90$. Half-spread $= 0.015(44.90) \approx 0.673$:

$$\text{bid} \approx \boxed{\$67.33}, \quad \text{ask} \approx \boxed{\$68.67}$$

▶ *Intuition:* identical pipeline to the chapter's Example 17, new

numbers — $\mathbb{E}[X] \to \mathbb{E}[X^2] \to \text{Var} \to \sigma \to$ spread. Running this four-step chain without hesitation, cold, is exactly what a Stage 1 timed question is testing.

Q10 — Capstone: Sequential Quoting, Fully Combined

Round 1: fair value \$90, flat, quote ±\$1 (89/91), filled on the bid for 3 units (now long 3). Round 2: fair value moves to \$92; $\gamma = 0.01$, $\sigma^2 = 25$. (a) Find the new reservation price and ±\$1 quote. (b) Using $\mathbb{P}(\text{fill}) = 0.5 - 0.25d$ at $d = 1$ on both sides, find net expected P&L for round 2 if holding cost is \$0.05/unit/round.

Solution. (a) $r = 92 - (3)(0.01)(25) = 92 - 0.75 = \boxed{\$91.25}$:

$$\text{bid} = \boxed{\$90.25}, \quad \text{ask} = \boxed{\$92.25}$$

(b) Both sides at $d = 1$: $\mathbb{P}(\text{fill}) = 0.5 - 0.25 = 0.25$ each, edge \$1 each:

$$\mathbb{E}[\text{P\&L}]_{\text{gross}} = 0.25(1) + 0.25(1) = 0.50$$

$$\text{Net} = 0.50 - 0.05(3) = 0.50 - 0.15 = \boxed{\$0.35}$$

⋆ *Full-circle intuition:* this one question chains every section of the chapter — an inventory-skewed reservation price (Section C), a symmetric quote built around it (Section B), and a fill-probability-weighted net P&L (Sections C+A's linearity) — in the same order a live multi-round market-making round actually unfolds.

Drill order for review: re-derive Q2 (explain why *it diverges, not just that it does), Q5 (spot a crossed market on sight), and Q10 (the full four-stage pipeline) closed-book. Together they test whether you can run fair value, quoting, and inventory logic as one connected skill, not three separate formulas.*